

Payment Gateway Timeout Outage - Card Present Transactions

Meridian Commercial Bank - Internal Document

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Summary

Between 09:12 and 10:47 UTC on 2025-08-14, card-present transaction authorizations routed through the Payments Gateway (PGW) experienced elevated latency and timeouts. Approximately 18% of authorization requests during the window failed with a gateway timeout (HTTP 504) rather than a normal approve/decline response.

Timeline

- 09:12 - PGW p99 latency alert fires (>4500ms, threshold 1200ms).
- 09:18 - On-call payments engineer acknowledges; begins triage using the Payment Gateway Failover runbook.
- 09:26 - Root cause suspected: connection pool exhaustion to the downstream card network adapter (CNA) service after a CNA deployment at 09:05.
- 09:40 - Decision made to fail over PGW traffic to the secondary region.
- 10:05 - Failover completed; latency begins recovering.
- 10:47 - Latency fully recovered; incident downgraded to monitoring.
- 11:30 - Incident closed.

Root Cause

The Card Network Adapter (CNA) service was redeployed at 09:05 UTC with a reduced connection pool size (default value was accidentally left at the framework default of 10 instead of the tuned production value of 200). Under normal peak load, PGW instances exhausted the available CNA connections, causing requests to queue and eventually time out at the PGW's 5-second upstream timeout.

Impact

- Approximately 18% of card-present authorization attempts failed for 95 minutes.
- Estimated 4,200 declined-in-error transactions.
- No funds were moved incorrectly; failures were pre-authorization.
- Merchant support received approximately 60 escalations.

Resolution

- Failed PGW traffic to secondary region (unaffected by the bad CNA config).
- CNA connection pool size restored to the tuned value of 200 and redeployed.
- Added a pre-deploy config diff check to the CNA deployment pipeline.

Action Items

1. Add automated regression test asserting CNA connection pool size before deploy (Owner: Payments Platform, Due: 2025-08-28).
2. Add PGW-to-CNA circuit breaker to fail fast instead of queuing under connection exhaustion (Owner: Payments Platform, Due: 2025-09-15).
3. Update the Payment Gateway Failover runbook with the CNA connection pool check as a first-line triage step (Owner: SRE, Due: 2025-08-21).